

```
from IPython.display import Video
import cv2
import os

# Path to the video file
video_path = '/content/beach.mp4'

if not os.path.exists(video_path):
    print(f'Error: {video_path} not found.')
else:
    cap = cv2.VideoCapture(video_path)
    fps = cap.get(cv2.CAP_PROP_FPS)
    width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

    # Limit to 3 seconds
    num_frames_to_process = int(3 * fps)
    frame_count = 0

    # Using 'avc1' (H.264) codec for better browser compatibility
    fourcc = cv2.VideoWriter_fourcc(*'avc1')
    slow = cv2.VideoWriter('slow_3s.mp4', fourcc, fps/2, (width, height))
    fast = cv2.VideoWriter('fast_3s.mp4', fourcc, fps*2, (width, height))

    while frame_count < num_frames_to_process:
        ret, frame = cap.read()
        if not ret:
            break
        slow.write(frame)
        fast.write(frame)
        frame_count += 1

    cap.release()
    slow.release()
    fast.release()
```

```
print('Slow Motion Video (3 Seconds):')  
display(Video('slow_3s.mp4', embed=True))  
print('Fast Motion Video (3 Seconds):')  
display(Video('fast_3s.mp4', embed=True))
```

Slow Motion Video (3 Seconds):

0:00



Fast Motion Video (3 Seconds):

0:00



